

Topologically constrained models capture functional aspects of a sensory code

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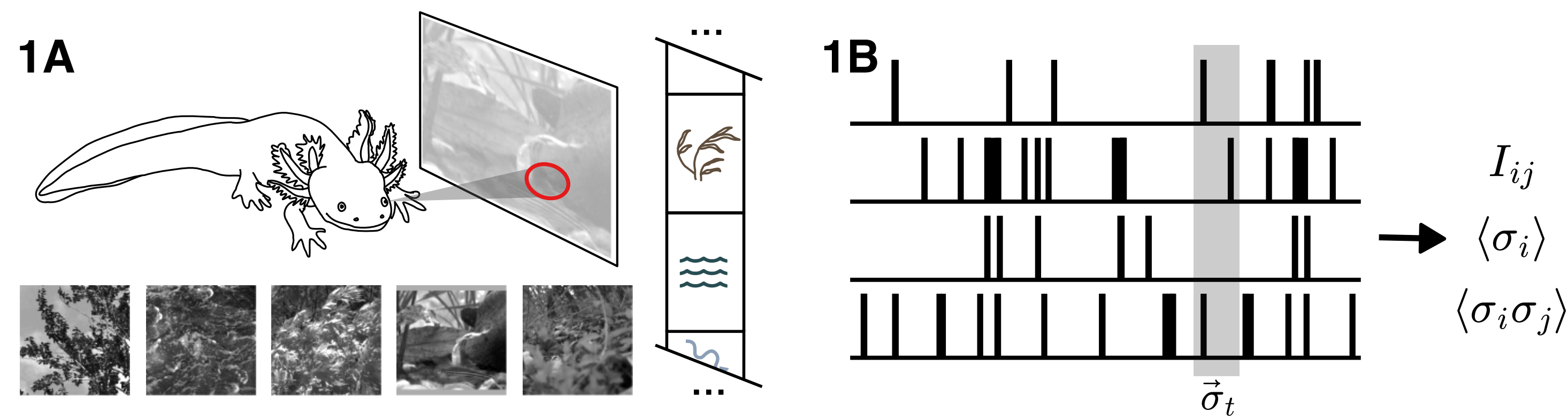


Background

The retina maintains a robust representation of complex visual environments through the coordinated activity of large populations of neurons, whose collective signals must be interpreted by downstream brain areas to guide behavior. **Minimax entropy models** [1] are a promising means of characterizing this activity, but previous analyses using these models have been largely agnostic to the downstream behavioral goals of an organism.

Methods

Response to dynamic natural scenes

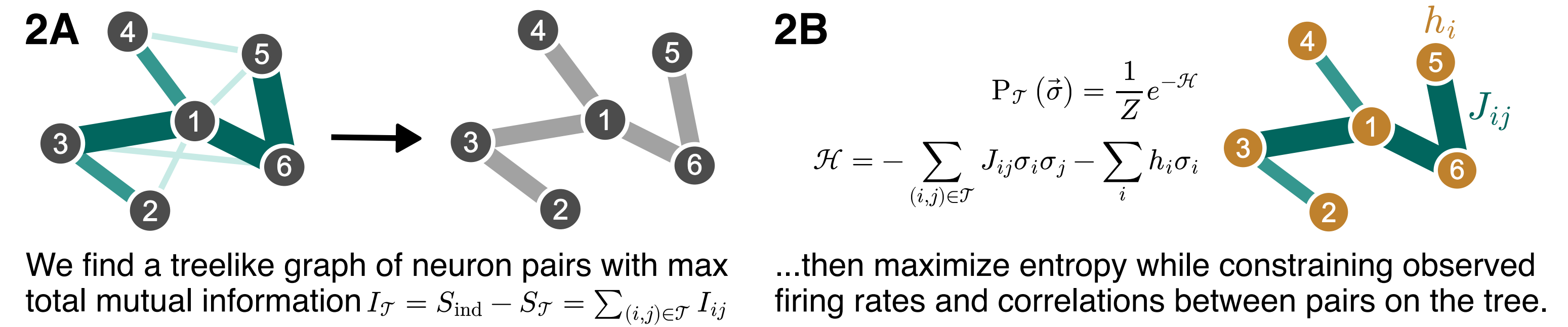


Tiger salamander RGC activity was recorded while the retina was presented with natural movies [2].

We measure average firing rates, pairwise correlations, and estimate cell-cell mutual information.

Methods

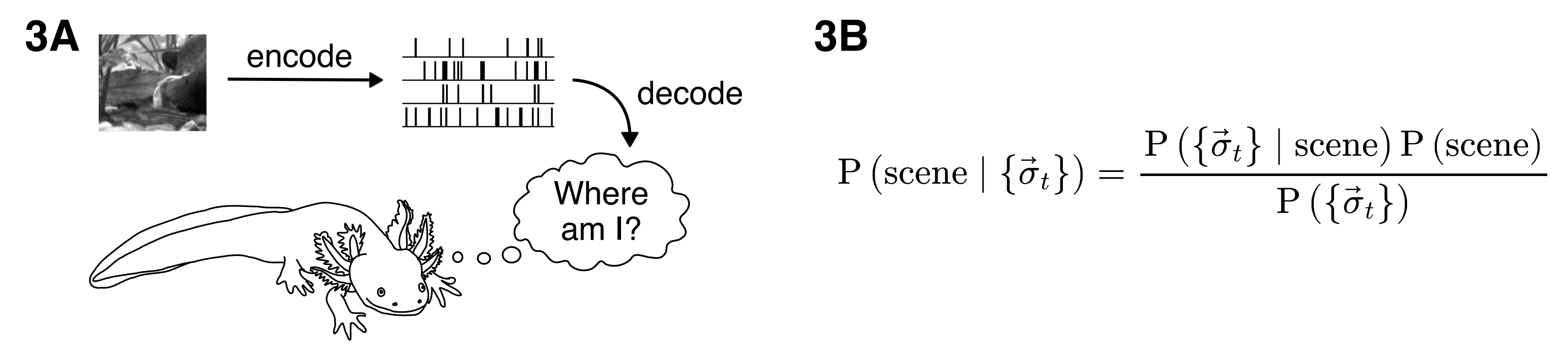
Minimax entropy model



We find a treelike graph of neuron pairs with max total mutual information $I_T = S_{\text{ind}} - S_T = \sum_{(i,j) \in T} I_{ij}$

...then maximize entropy while constraining observed firing rates and correlations between pairs on the tree.

Decoding scene identity

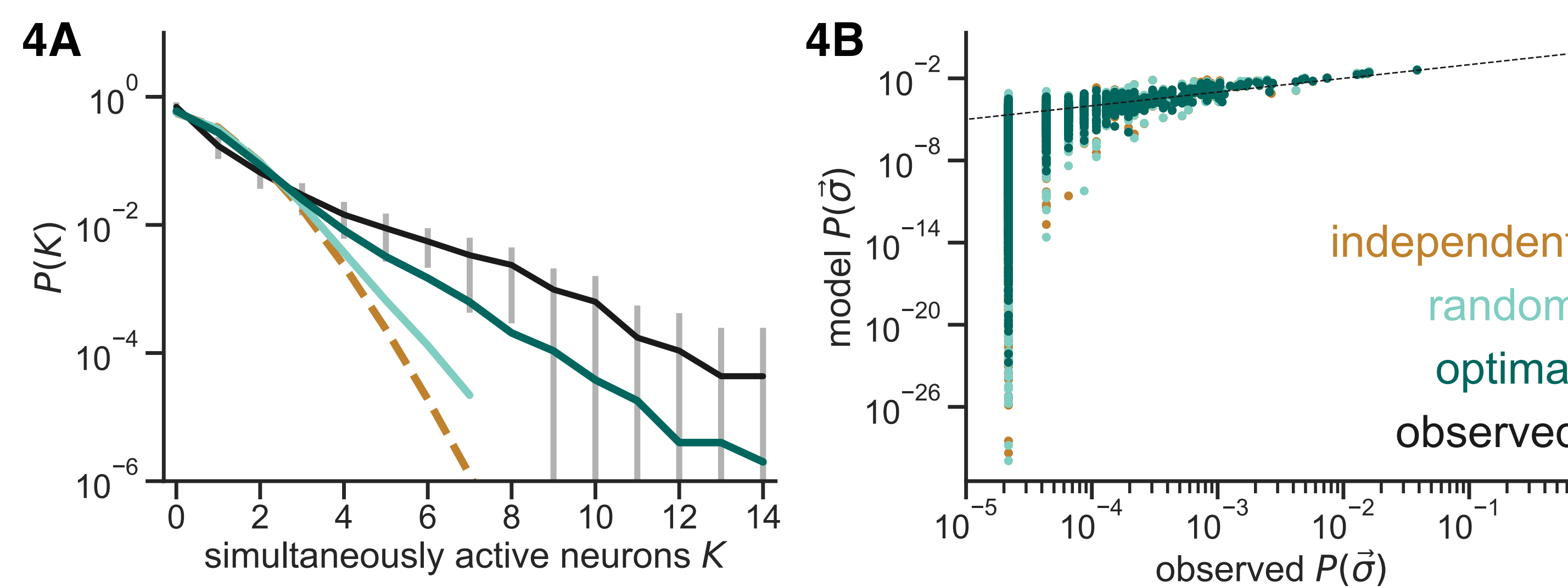


Broad scene structure is encoded in the retina and decoded in downstream brain areas to guide behavior.

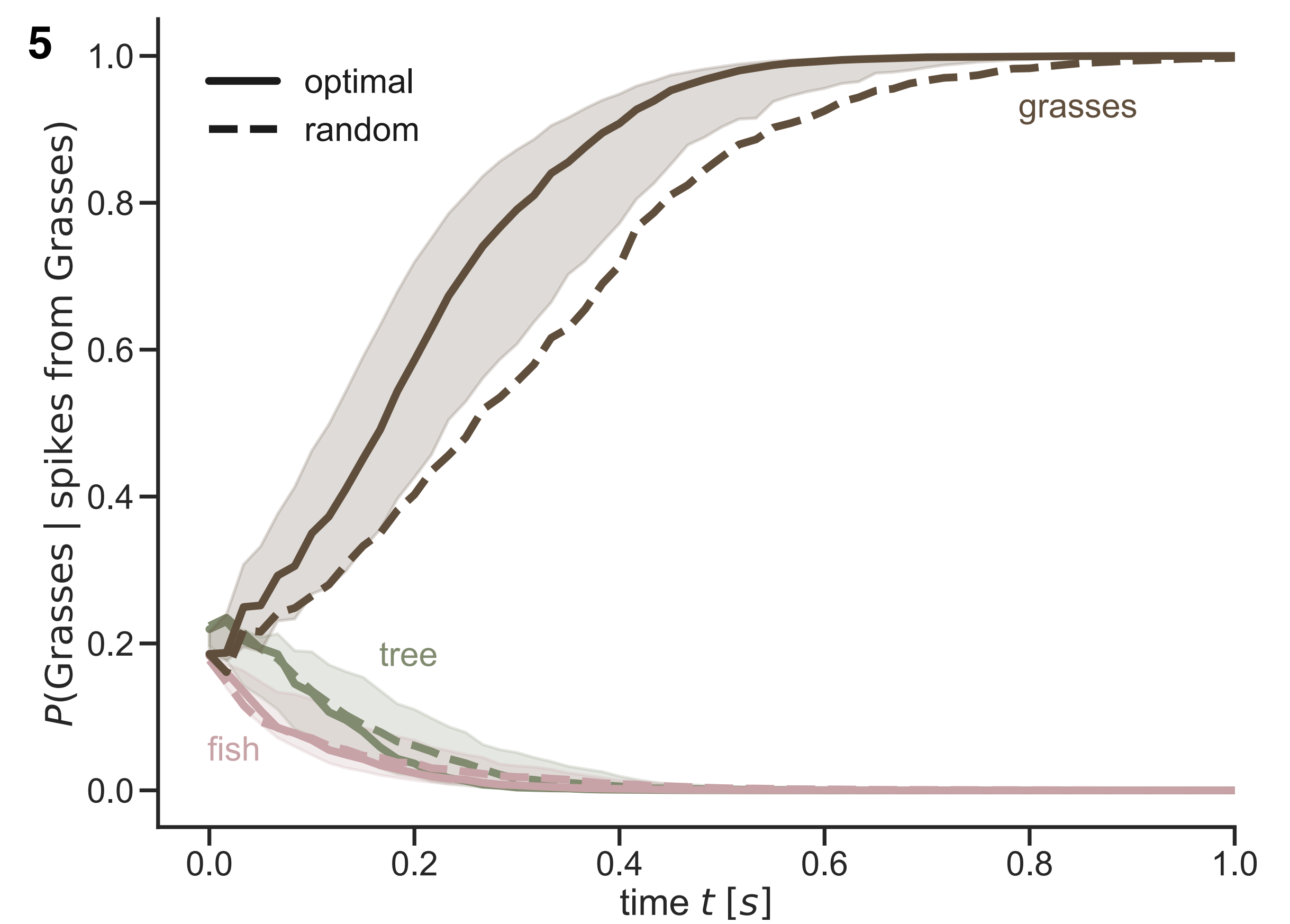
We evaluate scene identification performance via an ideal Bayesian decoder [3].

Results

Optimal, topologically constrained models consistently capture significantly more of the heavy-tailed structure of synchronous activity than both independent models and models with random tree interaction structures.



The optimal tree model also predicts the occurrence rates of common population response patterns to a high degree of accuracy



A sparse network of maximally informative couplings enables accurate and robust identification of natural scenes within a fraction of a second for all movies.

Conclusion

Topologically constrained, maximally informative coupling networks contribute to robust encoding of naturalistic stimuli and allow for efficient downstream decoding. Our results demonstrate the functional relevance of minimax entropy models in sensory codes, offering a computationally efficient and biologically interpretable statistical physics framework for understanding population coding in large neuronal ensembles.

a link to this poster →



References

- [1] C. W. Lynn, Q. Yu, R. Pang, S. E. Palmer, and W. Bialek, Exact minimax entropy models of large-scale neuronal activity, Physical Review E 111, 054411 (2025).
- [2] S. E. Palmer, C. Holmes, B. Hoshal, K. Bojanek, J. Salisbury, M. Berry, and O. Marre, Larval salamander retinal population data in response to natural movies from the Chicago Motion Database (2024), 10.5061/dryad.4qrj6qm8.
- [3] B. D. Hoshal, C. M. Holmes, K. Bojanek, J. M. Salisbury, M. J. Berry, O. Marre, and S. E. Palmer, Stimulus-invariant aspects of the retinal code drive discriminability of natural scenes, Proceedings of the National Academy of Sciences 121 (2024).

Acknowledgements

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Our model captures a backbone of predominantly large, positive coupling values between neurons. Despite sparse connectivity, average firing rates predicted by the optimal tree model deviate from the predictions of both random tree and independent models.